Fall 2026 CIS*3210
Assignment 1: HTTP and DNS Packet Analysis

Weight: 6%

Due on: October 5, 2026, at 11:59 PM

Purpose

In this assignment, you will use Wireshark to connect protocol fields to observable network behavior. You will analyze supplied traces containing DNS and unencrypted HTTP/1.0, HTTP/1.1, and HTTP/2 traffic, then create a small capture of your own interaction with a public HTTP server. Your explanations must use evidence from specific packets.

Learning outcomes

- Identify DNS queries and responses and interpret common DNS fields and resource records
- Relate DNS name resolution to the TCP connection and HTTP exchange that follow
- Interpret HTTP request and response headers in an unencrypted packet capture
- Compare connection management and message representation across HTTP/1.0, HTTP/1.1, and HTTP/2
- Use display filters, packet numbers, timestamps, ports, flags, and protocol fields as evidence
- Explain how caching and modern encrypted protocols can change what appears in a capture

Files provided

File	Contents
Trace_1_HTTP_with_DNS.pcap	DNS lookup, TCP connection, HTTP request and response
Trace_2_DNS_cases.pcap	A, AAAA, CNAME, and negative DNS responses
Trace_3_HTTP_1_0_Nonpersistent.pcap	HTTP/1.0 request, response, and connection closure
Trace_4_HTTP_2_Cleartext_Multiplexing.pcap	HTTP/2 setup, frames, streams, and multiplexed data

General instructions

- Answer all 30 questions in your own words. Complete every part of each question, include units, and show calculations where applicable.
- Support each technical answer with packet number(s) and the relevant Wireshark field or value.
- Use screenshots only when requested. Crop them to the relevant packet list or field and keep packet numbers visible.
- Do not capture logins, email, banking, health, or other private activity. Close unrelated applications before capturing.
- Do not scan, probe, automate repeated requests, or attempt to bypass any access control. Generate only the few requests required here.
- Use unencrypted HTTP only for this controlled exercise. Never submit credentials or personal information over HTTP.

Part A: Supplied HTTP and DNS trace

Open Trace_1_HTTP_with_DNS.pcap. Begin with the display filter **`dns || http || tcp.port == 80`**.

1. Identify the client, DNS resolver, and web server IPv4 addresses. State how you assigned each role. **[2 marks]**

2. Report the DNS query transaction ID, name, type, client source port, and resolver destination port. Identify the matching response and state the fields that link the two packets. **[2 marks]**

3. What IPv4 address and TTL are returned? Explain what the TTL controls. **[1 marks]**

4. Identify the three packets in the TCP three-way handshake and state the relevant TCP flags in each. Report the client and server TCP ports, and explain why the server uses a well-known port while the client uses an ephemeral port. **[4 marks]**

5. Report the HTTP request method, requested resource, HTTP version, and Host header. Give the response status code and reason phrase, and explain what the status means. **[3 marks]**

6. Report the Content-Type and Content-Length values. Verify whether the reassembled HTTP response body has the length stated in the Content-Length header. **[2 marks]**

7. Is the response carried in one TCP segment? Support your answer with packet numbers and TCP sequence information. **[1 marks]**

8. Calculate the time from the DNS query to its response and from the HTTP request to the first TCP segment carrying bytes of the HTTP response. Show both calculations and include units. **[2 marks]**

9. What evidence indicates that the HTTP/1.1 connection is intended to be persistent? Explain why the client's later decision to close the connection does not contradict that evidence. **[1 marks]**

10. Use Follow TCP Stream and state the application data visible to an observer. Explain why this visibility would differ with HTTPS. **[2 marks]**

Part B: Supplied DNS cases

Open Trace_2_DNS_cases.pcap and use dns as the initial display filter.

11. For each of the three DNS queries, report the queried name, query type, query packet number, and transaction ID in capture order. For the AAAA lookup, report the returned IPv6 address and identify the matching query. **[4 marks]**

12. For the first lookup, report the number of answers, distinguish the alias from the canonical name, and give the final IPv4 address. Explain why both answers are useful to the client. **[5 marks]**

13. Compare the UDP source and destination ports in one query-response pair. Explain why they reverse direction. **[2 marks]**

14. Interpret the RD and RA flags. What does each indicate about the client or resolver? **[3 marks]**

15. Identify the packet that reports the failed lookup. Give the response code and the Answer and Authority counts. Explain what NXDOMAIN means and why the response can still contain an SOA record. **[4 marks]**

16. Compare the TTLs of the positive answers with the SOA record TTL and SOA MINIMUM value in the negative response. State the effective negative-caching TTL and explain how these values can affect repeated DNS queries. **[2 marks]**

Part C: HTTP/1.0 nonpersistent connection

Open Trace_3_HTTP_1_0_Nonpersistent.pcap and begin with the display filter **`http || tcp.port == 80`**.

17. Identify the TCP handshake, HTTP request, HTTP response, and connection-closing packets. Report the request method, resource, HTTP version, and response status. State the relevant TCP flags in the handshake and closing packets. **[5 marks]**

18. What evidence shows that the connection is nonpersistent? Include both HTTP and TCP evidence. **[3 marks]**

19. Report the response Content-Length and verify the response-body length. **[2 marks]**

20. Compare this trace with Trace 1. Explain why the HTTP/1.0 request can omit the Host header but the HTTP/1.1 request cannot. Identify two other observed differences, and for each state whether it follows from the protocol version or is specific to these captures. **[5 marks]**

Part D: HTTP/2 frames and multiplexing

Open Trace_4_HTTP_2_Cleartext_Multiplexing.pcap. Begin with the display filter **`http2 || tcp.port == 80`**. Use **`http2`** alone when examining HTTP/2 frames. This controlled trace uses cleartext HTTP/2 with prior knowledge, also called h2c, so its frame structure can be examined without TLS keys.

21. Report the client and server IP addresses and TCP ports. Identify the packet containing the client connection preface and state the text with which the preface begins. **[2 marks]**

22. Identify the initial non-ACK SETTINGS frame sent by each endpoint and report the setting and value advertised by the server. Identify the SETTINGS ACK sent by each endpoint. How is an acknowledgement represented in a SETTINGS frame? **[5 marks]**

23. Identify the two request stream IDs and the requested paths. Why are the client-initiated stream IDs odd? **[3 marks]**

24. State the purposes of the HEADERS and DATA frame types in the trace. List the DATA frames in capture order, giving each packet number, stream ID, and END_STREAM status. Explain how their ordering demonstrates multiplexing. **[6 marks]**

25. Compare HTTP/2 with the HTTP/1.x traces in message format, stream use, and connection use. **[4 marks]**

Part E: Live capture of name resolution and HTTP

Optional DNS cache clearing: On a personal computer, you may clear the operating system's DNS cache using the appropriate command below. Do not run a command with administrator privileges or **`sudo`** on a shared or institution-managed computer unless you are authorized to do so. If you cannot clear the cache, choose a hostname you have not recently accessed and document this limitation. Clearing the local cache does not clear upstream resolver caches and does not guarantee that ordinary DNS packets will appear.

Create one short capture that begins before name resolution and ends immediately after one HTTP response. Use a wired or Wi-Fi interface that carries your normal Internet traffic. Do not use a loopback capture.

Step 1: Attempt to produce a DNS cache miss

Choose one well-known public site that you have not accessed during the current session: **http://example.com/** or **http://www.google.com/**. If both hostnames may already be cached, clear the operating-system DNS cache using the applicable command or command sequence below. If you cannot run an administrative command, choose the less recently used hostname and document that limitation.

System	Command
Windows (run as Administrator)	`ipconfig /flushdns`
macOS (Terminal)	`sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder`
Linux with systemd-resolved (Terminal)	`sudo resolvectl flush-caches`

Step 2: Capture the exchange

- Close unrelated applications and browser tabs, then start Wireshark on the active interface.
- Run one command for the selected hostname:
  Windows: `curl.exe --http1.1 http://example.com/`
  macOS or Linux: `curl --http1.1 http://example.com/`
  Use the command exactly as shown. Do not add the **`-L`** option because the initial HTTP redirect, if present, is part of the traffic you must analyze.

  To use Google instead, replace `example.com` with `www.google.com`.
  Windows note: Type `curl.exe` exactly as shown.

- Stop the capture as soon as the response is complete. Save the unmodified file as **A1_<StudentNumber>_<LastName>.pcapng.**
- Apply **dns || http || tcp.port == 80** as a display filter. A display filter does not remove packets from the saved capture.

If no ordinary DNS packets appear, record what you observed. Start a new replacement capture and run **`nslookup`** for the selected hostname. Without stopping the capture, run the same **`curl`** command again. Stop the capture after the HTTP response and submit this replacement capture as **`A1_<StudentNumber>_<LastName>.pcapng`**.

If **`nslookup`** was required, you may compare its returned addresses with the destination address of the HTTP TCP connection. However, describe this as an observational comparison; the **`nslookup`** result is not necessarily the DNS result used internally by **`curl`**.

A 3xx HTTP redirect response is a valid live result and should be analyzed in Question 29. If the selected host is blocked or intercepted and no usable HTTP response is captured, try the other approved hostname in the same capture. Use the supplied-trace fallback only if neither host produces a usable HTTP response. If neither host works, save the capture showing the failed attempts and document the network restriction. In Question 28, identify the SYN attempt and explain that an RTT cannot be calculated if no SYN-ACK was received. Use Trace 1 for the HTTP field-identification portion of Question 29. In Question 30, compare the failed live attempt with the successful supplied trace. A documented network restriction will not be penalized.

26. State the selected hostname, operating system, capture interface, and whether you cleared the DNS cache. **[1 mark]**

27. Identify all relevant DNS query and matching response packet numbers. Report the query type or types, resolver IP address, response code or codes, and returned IP addresses. For the TCP stream carrying the HTTP request, determine whether its destination address appears in the captured DNS response. If you used the **`nslookup`** fallback, describe the comparison as observational and explain any mismatch. Include one cropped screenshot showing the DNS evidence. **[6 marks]**

28. Identify the three-way handshake for the TCP stream carrying the HTTP request. If multiple connection attempts appear, identify the successful stream and briefly account for the other attempts. Calculate the approximate RTT from the matching SYN and SYN-ACK timestamps. **[2 marks]**

29. Report the HTTP request method, Host header, destination port, and response status. Report and explain any HTTP redirect. Include one cropped screenshot showing the relevant request or response fields. **[3 marks]**

30. Compare your live capture with one supplied trace. Explain two meaningful similarities or differences and relate each observation to the server, client, cache, resolver, or network, as appropriate. **[3 marks]**

Submission requirements

- Submit one PDF report named according to this convention: **`A1_<StudentNumber>_<LastName>.pdf`** (Replace the placeholders with your information and do not include the angle brackets.)
- Organize the report by Parts A through E. Keep the question numbers shown in this handout.
- Place each screenshot next to the answer it supports. Packet numbers and relevant field values must be readable.
- Prepare the report using Microsoft Word or another word processor and export it to PDF. Do not submit a scanned handwritten report or an image-only PDF.
- Submit the original live-capture file named according to this convention:
  `A1_<StudentNumber>_<LastName>.pcapng` (Replace the placeholders with your information and do not include the angle brackets.)
- Remove unrelated or sensitive traffic by repeating the controlled capture, not by editing packet contents.
- Upload the PDF report and the live-capture file together to the designated Assignment Dropbox on CourseLink.
- Upload both required files in the same submission. Only the most recent submission will be graded, and it must contain both the PDF report and the live-capture file. If you resubmit either file, include both files again.
- Incorrect filenames will be assessed under Communication and Submission Quality. If the capture file is missing or cannot be opened, marks requiring verification from that file cannot be awarded. If the PDF report is missing or cannot be opened, the assignment cannot be graded.

Grading Scheme

This assignment is graded out of **100 marks**.

Criterion	Marks	What is assessed
Part A HTTP/1.1 and DNS	20	DNS, TCP, and HTTP/1.1 fields; supported calculations
Part B DNS cases	20	DNS records, flags, response codes, and caching
Part C HTTP/1.0	15	Nonpersistent behavior and comparison with HTTP/1.1
Part D HTTP/2	20	Frames, settings, streams, and multiplexing
Part E live capture	15	Controlled capture, evidence, and comparison
Communication	10	Clear organization, concise explanations, readable evidence, and correct filenames
Total	100	Assignment grade; worth 6% of the final course grade

How Answers Will Be Graded

Marks within each question will be awarded for:

- Correct identification and interpretation of protocol fields
- Appropriate packet or frame numbers
- Relevant evidence from the Wireshark trace
- Correct calculations, including units
- Clear explanations using appropriate networking terminology
- Readable and relevant screenshots where requested

Partial Credit

Partial marks may be awarded when an answer demonstrates correct reasoning or identifies some of the required fields but is incomplete or contains a minor error.

A calculation may receive method marks when the correct timestamps and calculation procedure are shown, even if the final value contains a small arithmetic error.

If an answer depends on an earlier incorrect value, later work may still receive marks when the method is correct and the answer is internally consistent.

Live-Capture Results

Results in Part E may differ among students because of:

- IPv4 or IPv6 use
- DNS caching or encrypted DNS
- Content delivery networks
- Proxies or VPNs
- HTTP redirects
- Institutional or local network restrictions
- Differences in DNS responses and server addresses

Students will be graded on the accuracy of their observations, evidence, and interpretation and not on obtaining a specific IP address, status code, or timing value.

If the permitted **`nslookup`** fallback is required, run **`nslookup`** and **`curl`** in the same replacement capture and submit that capture using the required filename.

Evidence and Screenshots

- Screenshots must be readable and show the relevant packet details.
- Screenshots support, but do not replace, written answers.
- Packet numbers and important field values should be stated in the report.
- The submitted capture file must allow the reported observations to be verified.
- If a capture file is missing or cannot be opened, marks requiring verification from that file cannot be awarded.

Communication and Submission Quality (10 Marks)

Criterion	Marks
Report organization and correct question numbering	1
Clear explanations and appropriate terminology	1
Clearly presented packet references, values, calculations, and units	1
Relevant, cropped, and readable screenshots	1
PDF report follows the required filename convention	2
Capture file follows the required filename convention	2
Student number and last name are complete and consistent in both filenames	1
Both required files are included, openable, and submitted together	1
Total	**10**